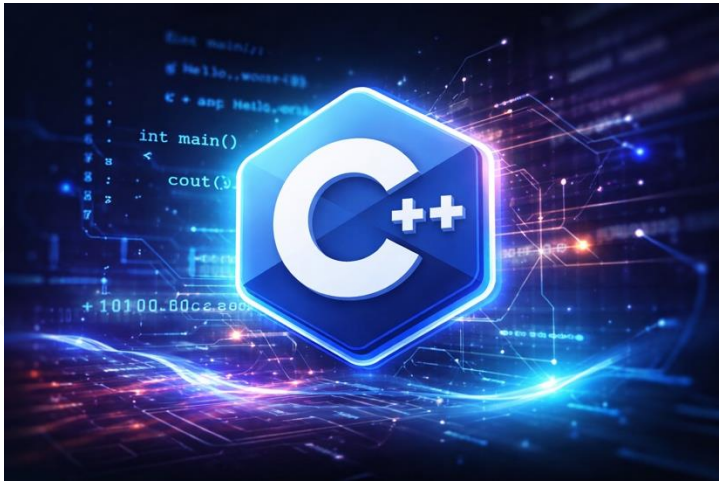
WEEK 1: COURSE INTRODUCTION & YOUR FIRST C++ PROGRAM



CSCI271 Problem Solving and Programming 1
Modesto Junior College
Wendy Chen

Wendy Chen

Overview

- Welcome & Course Introduction
  - How to Get Support
  - Course Overview
- What is a Computer?
  - Memory Hierarchy: The Big Picture
  - How a computer runs a program
  - The Software Development Method
  - Why C++?
  - What is IDEs
  - Popular IDEs for C++
- GitHub Setup
- What To Do in Week 1

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Welcome & Course Introduction

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How to Get Support – Stay Connected

- **Canvas Inbox**
  - Send a Canvas Inbox message with a clear **topic**.

- **Email: chens@yosemite.edu**
  - Send email with: CSCI271-section [Topic]

- **Virtual Office Hours**
  - Details in the syllabus

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Course Overview-1

- **What is this course?**
  - The *first core course* for Computer Science majors and transfer students at MJC.
  - No previous C++ experience required, but you should have basic programming skills (from CSCI 270 or equivalent).
  - We will learn how to solve problems by designing, writing, and debugging programs using the C++ language.

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Course Overview-2

- **What Will You Learn?**
  - Understand and apply object-oriented programming (OOP) concepts: classes, objects, encapsulation, inheritance, and polymorphism.
  - Design and implement solutions using algorithmic thinking and structured problem-solving.
  - Understand and use good software engineering practices (style, documentation, modularity).
  - Complete assignment that model real-world scenarios.

Working Open

How Does This Online Course Work?

- All lectures, assignments, and materials are on Canvas.
  - You are expected to check Canvas regularly, follow due dates, participate in discussion forums.
- No scheduled Zoom meetings; you can learn at your own pace each week.
- Weekly modules: Readings, videos, quizzes, lab assignments, programming projects.
- Distance Lab: all labs and programming projects completed online.

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How will you be graded in MCSCI-271?

Component	Weight	Description
Weekly Lab	**35%**	**Programming exercises submitted each week, based on topics from lectures and readings**
Assignments	**20%**	**Homework assignments are designed to deepen your understanding of the topics covered in class. Each homework assignment includes multiple-choice and/or true/false questions and allows up to two submission attempts. Correct answers will be released after Wednesday each week.**
Quizzes	**15%**	**Short quizzes and/or unit tests covering key concepts**
Reflections/Participation	**10%**	**Involvement in Canvas discussions, peer review, asking questions, and sharing ideas**
Midterm and Final Exams	**20%**	**Comprehensive programming project or cumulative final exam**

Grading scale: A (90–100%), B (80–89%), C (70–79%), D (60–69%), F (<60%). All details are in the syllabus on Canvas.

Other Policies

- **Late assignments:** refer to syllabus
- **Academic Integrity:** All work must be your own unless otherwise specified. Plagiarism or copying code is a serious violation.
- **Collaboration:** Discussion and idea sharing is encouraged, but all submitted code must be written by you.

Work Open

Tips for Success

- Stay on top of weekly tasks; don’t fall behind!
- Ask questions early — Canvas discussion board is always open.
- If you get stuck, reach out to Wendy or classmates for help.
- Extra resources: textbook, coding tutorials, recommended websites.

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Textbook and Computer Requirements

- *C++ Programming: Program Design Including Data Structures, 8th Edition, MindTap Course List, by D.S. Malik*
- This textbook is required for CSCI 271 and will also be used in CSCI 272. A 1-Year Subscription is recommended, since it will cover both courses.

- Need access to a Windows, macOS, or Linux computer to run modern development environments.
- **Chromebooks are NOT suitable for this course** due to limitations

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What is a Computer?

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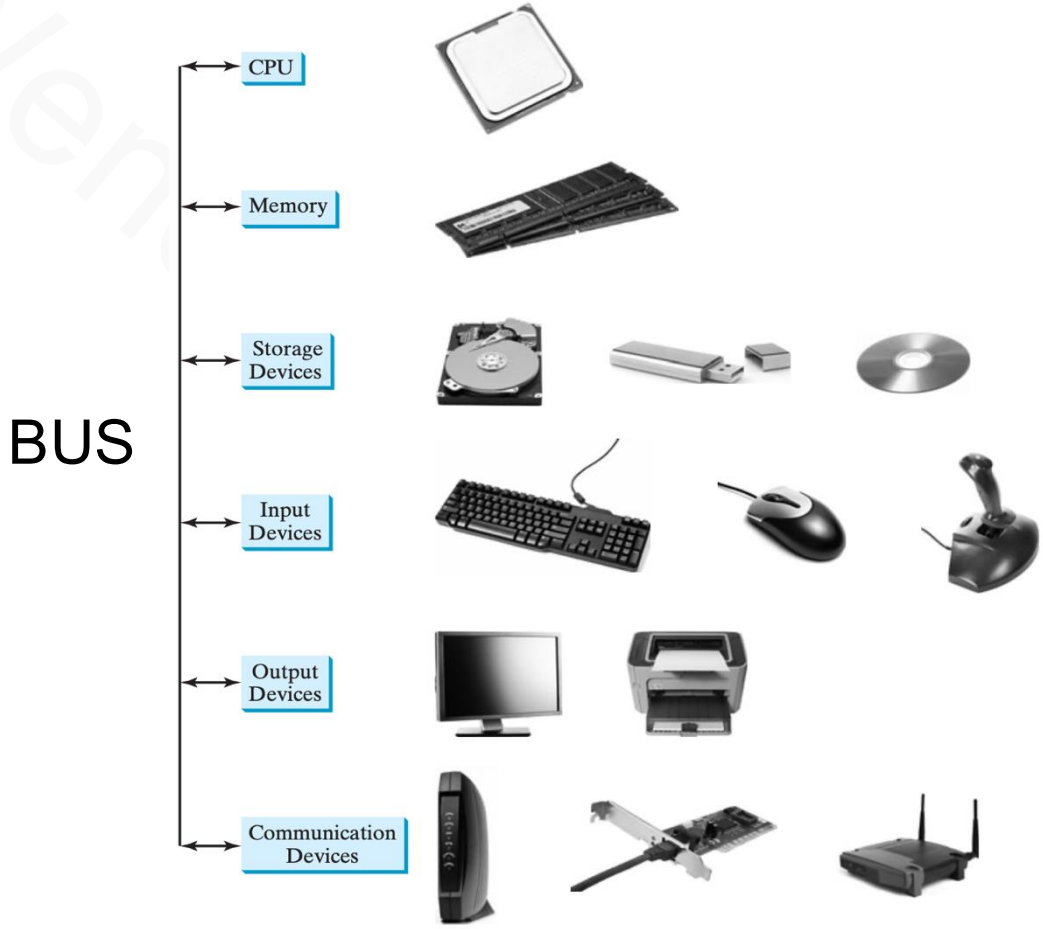
What is a Computer?

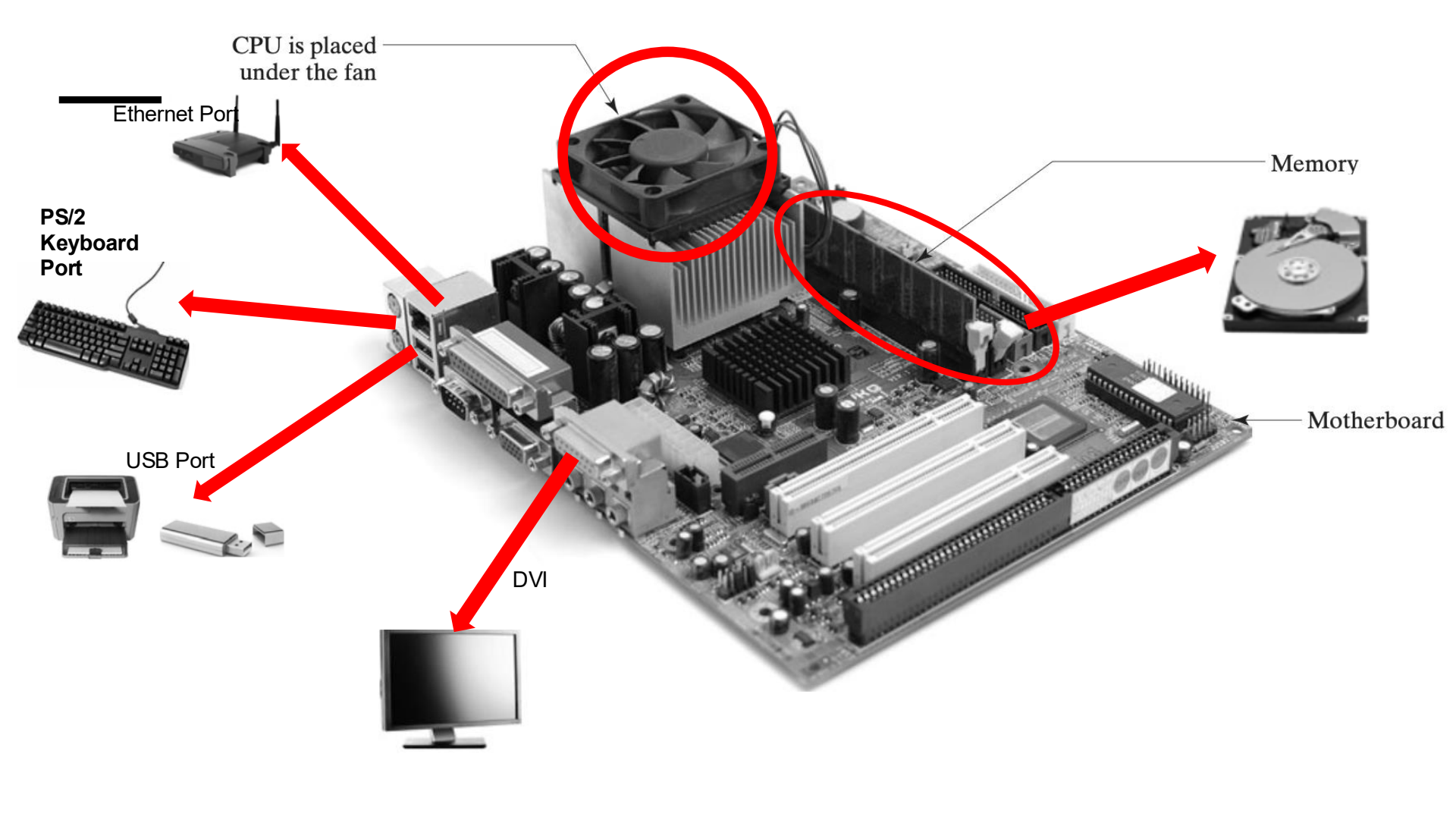
- **A computer** is an electronic device that processes and stores data, and can perform tasks based on a set of instructions called a program.

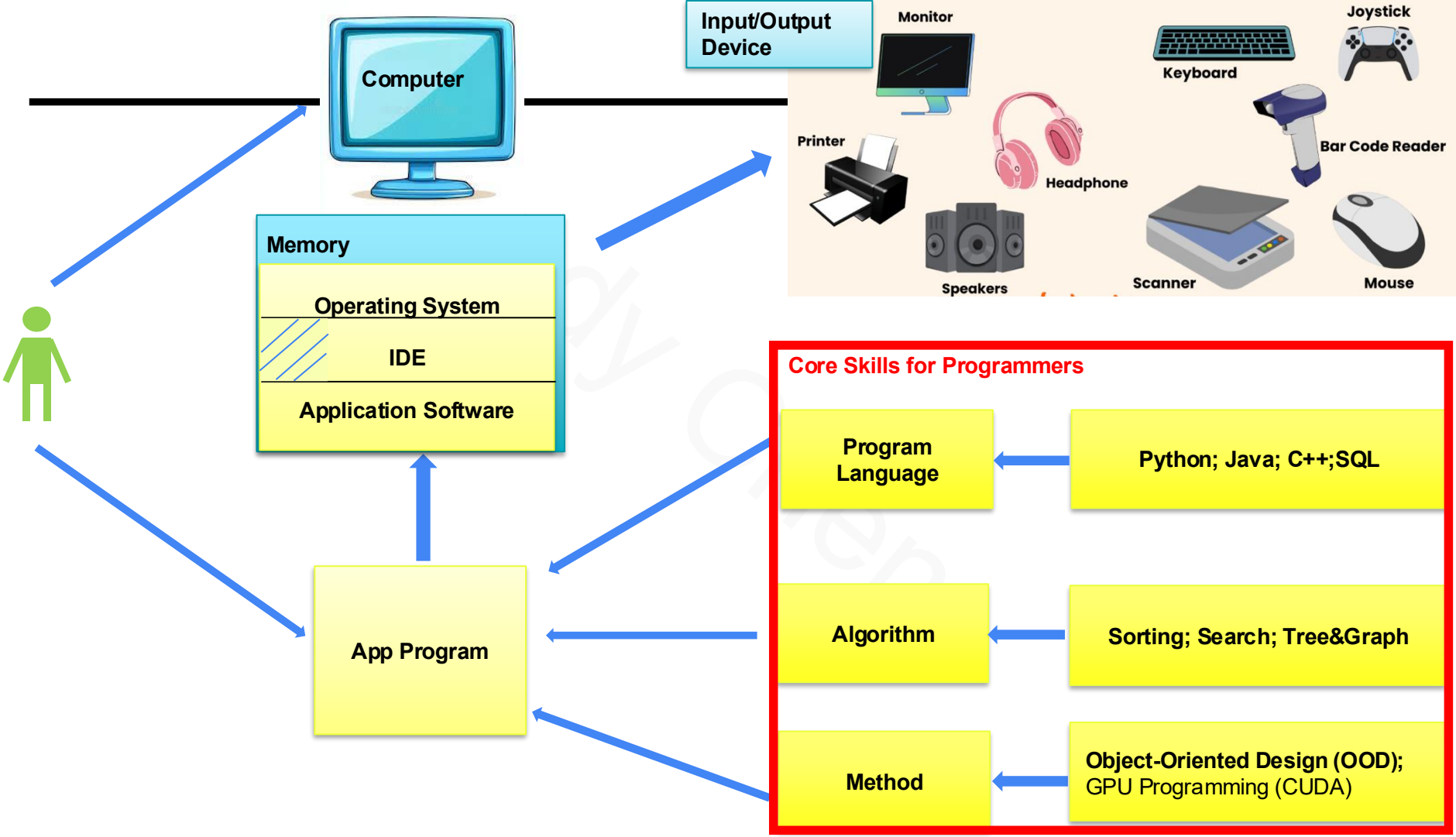
- Main components:
  - **CPU (Central Processing Unit):** The “brain” of the computer, processes instructions.
  - **Memory (RAM):** Temporarily stores data and instructions.
  - **Storage:** Hard drive, SSD, etc., for long-term data storage.
  - **Input Devices:** Keyboard, mouse, etc.
  - **Output Devices:** Monitor, printer, speakers, etc.
- Reference: *Connecting with Computing Science*, Ch. 1

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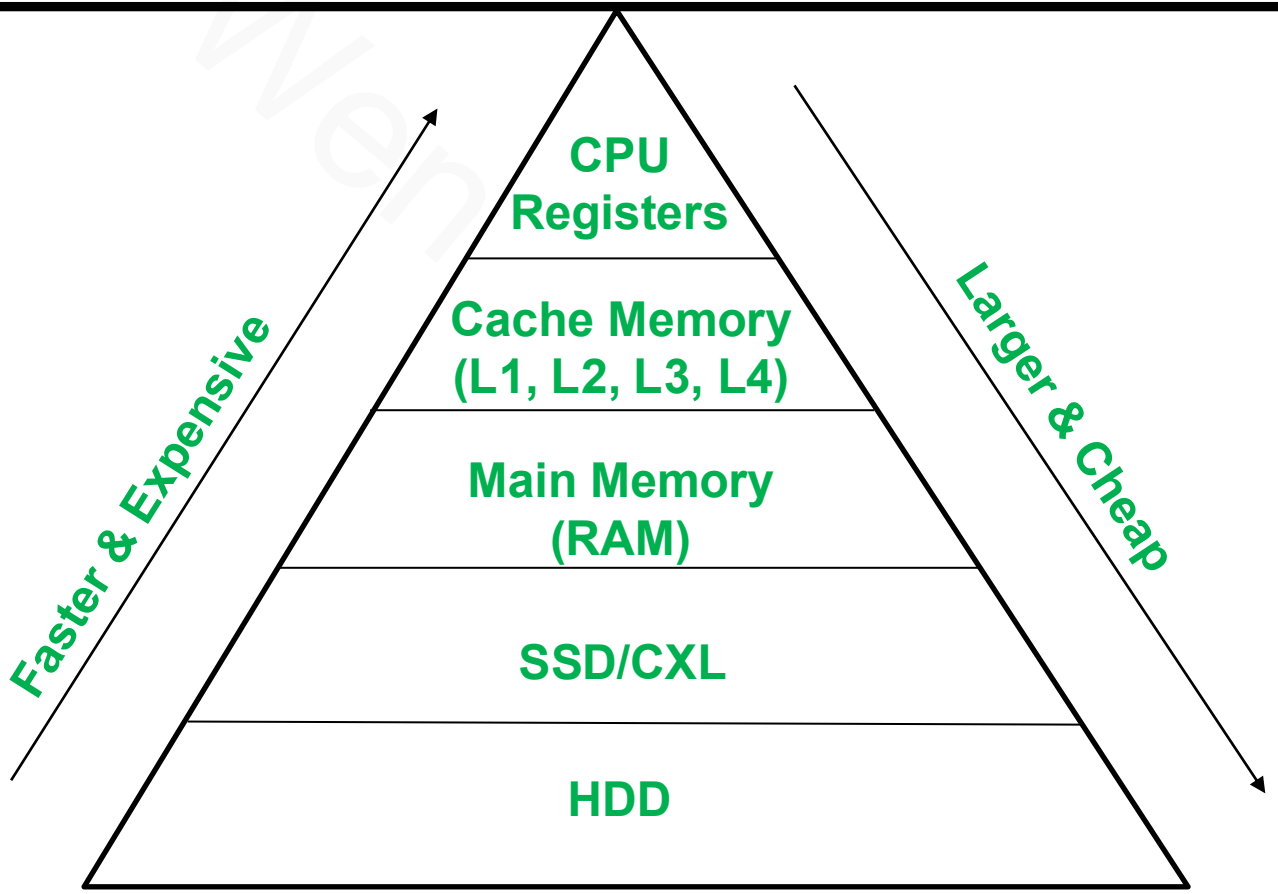
COMPUTER


BUS
CPU
Memory
Storage Devices
Input Devices
Output Devices
Communication Devices

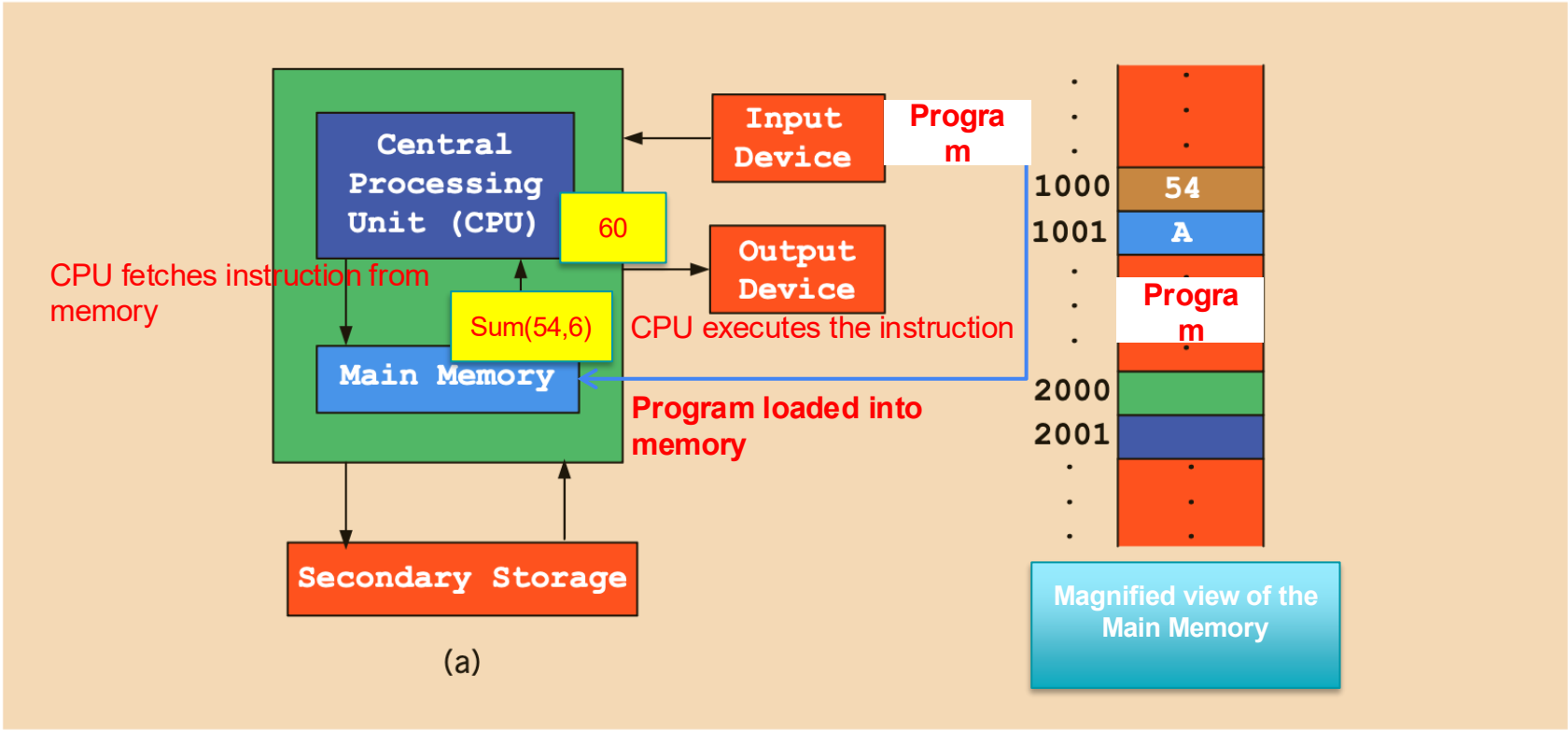

CPU is placed under the fan
Ethernet Port
PS/2 Keyboard Port
USB Port
DVI
Memory
Motherboard


Computer
Input/Output Device
Monitor
Joystick
Keyboard
Printer
Headphone
Bar Code Reader
Speakers
Scanner
Mouse
Memory
Operating System
IDE
Application Software
App Program
Core Skills for Programmers
Program Language
Python; Java; C++;SQL
Algorithm
Sorting; Search; Tree&Graph
Method
Object-Oriented Design (OOD);
GPU Programming (CUDA)

Memory Hierarchy: The Big Picture

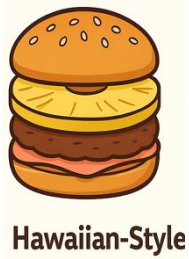


How a computer runs a program





Flavor









Write(Cook)

Write Instructions (Program)

- Place the bread
- Add the patty
- Pan-Fry for 5 minutes
- Add Pineapple
- Serve

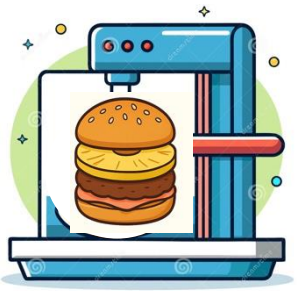
Method

- Grill
- Bake
- Pan-Fry
- Deep-Fry
- Steam

Language

- English
- Spanish
- French

Burger machine follows the instructions automatically!



Burger Machine

Burger Machine Analogy for Programming 

- **Key Roles**
  - **Burger Machine (Multi-Function Cooker) → Computer (Hardware + System)**
  - **Function Keys (Grill, Pan-fry, Deep-fry, Steam, Bake) → Method** (Specific ways to perform actions)
  - **Instruction Manual (Step-by-step) → Program** (A set of instructions to follow)
  - **Recipe Flavor (Italian, German, Mexican, Hawaiian) → Algorithm** (Different ways to achieve the goal)
  - **Chef's Language (English, Spanish) → Programming Language** (How the steps are expressed)

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Burger Machine Analogy for Programming 

Programming Styles:

- **Object-Oriented Design (OOD)** → Treats ingredients, tools, and methods as "objects" (e.g., a "Bread" object, an "Oven" object).
    - *Example:* Like a burger recipe where "Bread" has its own settings and "Oven" has its own functions.

- **Structured Programming Design**→ Step-by-step instructions without grouping into objects.
    - *Example:* A plain recipe:
        1. Put bread in oven.
        2. Bake for 5 minutes.
        3. Add cheese.

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The Software Development Method

- Software development follows **six key steps**:
  1. Clearly define the problem.
  2. Plan a solution (algorithm, flowchart, or pseudocode).
  3. Implement the solution (write code).
  4. Test and debug the program.
  5. Document the program (add comments and explanation).
  6. Maintain and update as needed.

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Why C++? & A Brief History

- **C++** is widely used for systems, applications, games, and more.
- Combines the power of C (performance, low-level control) with OOP features.
- Created in the early 1980s.
- **Standardized** by ANSI/ISO; has a large standard library.

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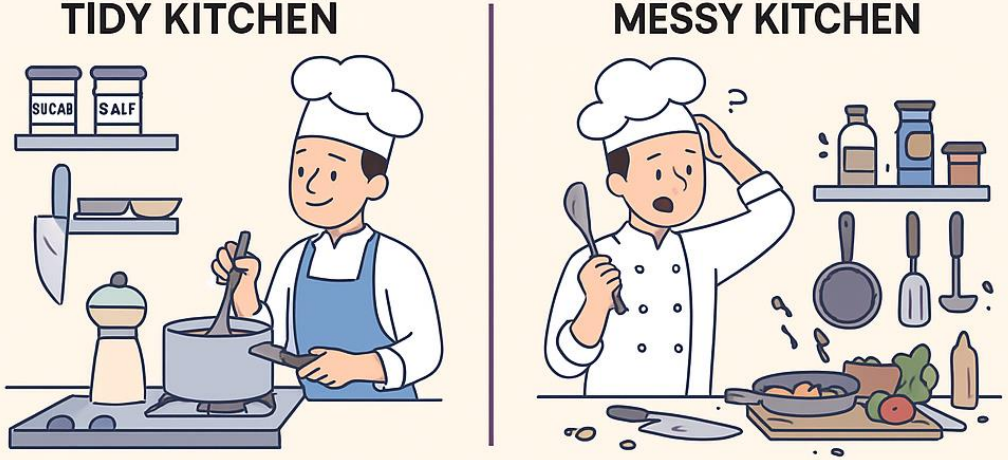
Good Programming Style & Using Online Resources

- **Programming Style:**
  - Use meaningful names for variables and functions.
  - Add comments to explain your code.
  - Keep code organized and readable (indentation!).
- **Documentation:**
  - Helps you and others understand and maintain your code.
- **Online Resources:**
  - Use tutorials, coding forums, and documentation (Cplusplus.com, StackOverflow, etc.)
  - The internet is your friend when learning to code!

Keep Open

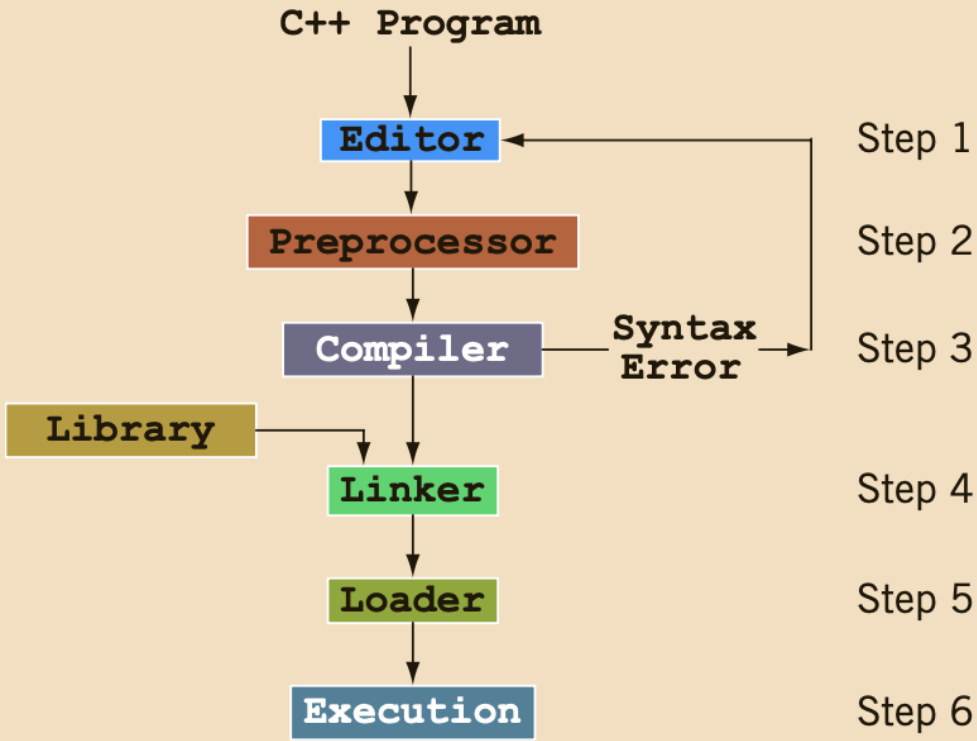
Good Programming Style

THINK OF YOUR CODE LIKE A KITCHEN



- Use meaningful **names** for variables and functions.
- Add **comments** to explain your code.
- Keep code **organized** and readable (indentation!).
- Documentation helps you and others.
- Use **online resources** (Cppluslus.com, StackOverflow, etc

How a typical C++ program is processed



IDEs are quite user friendly

- What is an IDE?
  - Integrated Development Environment = all-in-one coding tool
  - Includes: **Editor**, **Compiler/Debugger**, **Build Tools**, **Project Manager**
- Compiler helps identify syntax errors and suggests fixes
- Build or Rebuild links object code with project resources

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Popular IDEs for C++

- **Visual Studio 2022 Community (Windows Only)** – easy installation, very powerful
  - Download: https://visualstudio.microsoft.com/vs/community/
  - Installation Run Video: https://www.youtube.com/watch?v=7dXfMowQIfE.
- **VS Code (Cross-Platform)** – lightweight, but needs separate compiler
  - Download: https://code.visualstudio.com/download
  - Installation Run Video **for Windows**: https://www.youtube.com/watch?v=DMWD7wfhgNY
  - Installation Run Video **for MacOS**: https://www.youtube.com/watch?v=tdAD0WZjXrM
  - If you choose VS Code, you will need to install a C++ compiler separately (MinGW for Windows, GCC for Linux/macOS). The details shows in the Installation Run Video
- **CLion (All OS, Recommended)** – **free for students**, full features, cross-platform
  - Download: https://www.jetbrains.com/clion/
  - Installation Run Video: https://www.youtube.com/watch?v=HSf-GiJr1Bs
- **Xcode (MacOS Only)** – free from Mac App Store
  - Download: https://developer.apple.com/xcode/
  - Installation Run Video: https://www.youtube.com/watch?v=HA-NahwEnyA

Processing a C++ Program

```
#include <iostream>
using namespace std;
int main()
{
        cout << "My first C++ program." << endl;
        return 0;
}
```

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Run (execute) this program

My first C++ program.

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Week 1: What To Do -1

- **Log in to Canvas and explore the course site**
  - Find announcements, weekly modules, and assignment deadlines

- **Read the syllabus**
  - Understand course goals, grading, late policy, and participation requirements

- **Read the textbook:**
  - **Malik, Ch.1 (Introduction to Computer Science and C++)**
    (Required: Get an overview of CS, C++ history, and the big picture of programming)
  - **Malik, Ch.2 (Getting Started with C++) — Sections on Hello World & program structure**
    (Optional: Focus on understanding the basic structure of a C++ program and how to run it. You can review the rest of Ch.2 next week.)

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Week 1: What To Do -2

- **Take the Student Information/Technology survey**
- **Take the Week 1 Assignment**
  - Questions about basic and run your first c++
- **Finish the Lab 1 (GitHub setup)**
- **Setup your IDE Environment by watching the Installation Run Video**
  - Share a Screenshot of the running result under **Week 1 Assignment**

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